Real-Time Ransomware Detection Using Hardware Performance Indicators

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***Abstract*—Ransomware is arguably one of the most destructive types of malwares that has resulted in billions of dollars in worldwide financial loss each year, particularly in critical infrastructures, healthcare, and business environments. Signature detection and application layer behavioral detection methods are no longer effective against various forms of polymorphic ransomware attacks that have been developed over time. This paper proposes a real-time ransomware detection framework that utilizes machine learning classifiers based on Hardware Performance Counters (HPCs), which are low-level micro-architectural signals of the central processing units of computers. As studied in [1], [6], [7], the K-LEB kernel-level collection tool was used to collect HPC data every 10 ms for 12 variants of ransomware attacks: ALPHV/BlackCat, REvil, Sodinokibi, HelloKitty, Monti, RansomEXX, HolyCrypt, TellYouThePass, Golang, CryptSky, C-variant, and another variant of BlackCat ransomware, alongside a benign sample, resulting in a total of 285,871 raw data points for six hardware events: branch instructions retired (BR_RET), instructions retired (INST_RET), data cache accesses (DCACHE_ACCESS), load dispatches (LOAD), store dispatches (STORE), and L3 cache misses (MISS_LLC). A sliding window segmentation scheme is used for feature extraction with a window size $W = 50$ at a resolution of 10 ms, resulting in 500 ms observation windows. Two machine learning classifiers are evaluated: Random Forest as the primary classifier and Support Vector Machine (SVM) as a baseline. Random Forest is selected for its ensemble-based robustness, built-in feature importance estimation, and superior generalization on high-dimensional tabular data. The Random Forest achieves 97.45% accuracy, 98.20% precision, 95.96% recall, and F1=0.9707, substantially outperforming the SVM (85.7% accuracy, F1=0.8232) across all metrics. Random Forest feature importance analysis identifies STORE dispatches (31.45%) as the dominant ransomware signal, followed by data cache accesses (19.48%) and instruction retirement rate (15.08%), providing interpretable hardware-level evidence of encryption activity.**

***Index Terms*—Hardware Performance Counters; K-LEB; Ransomware Detection; Random Forest; Support Vector Machine; SVM; Sliding Window; Real-Time Monitoring; Behavioral Detection; Cybersecurity**

I. Introduction

In the current threat landscape, ransomware has become the most economically damaging type of malware. Ransomware is one of the top three action types in confirmed breaches worldwide, according to the 2023 Verizon Data Breach Investigations Report, and median ransom payments are still rising annually [1]. Significant events, such as the shutdown of the Colonial Pipeline in 2021, the declaration of a national emergency in Costa Rica in 2022, and the attack on MGM Resorts in 2023, show that government services, big businesses, and vital infrastructure are all high-value targets. Mass file encryption is the main effect of ransomware; by encrypting a victim's data using symmetric ciphers such as AES-256, ChaCha20, or Salsa20 and demanding payment for the decryption key [2], attackers cause instantaneous operational paralysis across all industries.

The effectiveness of traditional defenses against contemporary ransomware families is declining. Because every infection modifies its own binary signature, signature-based antivirus engines stop working within hours of the deployment of a new polymorphic variant—a fundamental boundary of static analysis first rigorously characterized by Moser et al. [3]. Automated dynamic analysis pipelines offer greater coverage by executing samples in instrumented environments [4], but they are compromised by sandbox-aware dormancy, in which malware postpones execution until it concludes it is not operating in an analysis environment. Graph-based behavioral profiling [5] captures richer call-graph anomalies yet remains vulnerable to obfuscation and imposes runtime overhead that is prohibitive in latency-sensitive deployments. In order to evade rule-based behavioral monitors, living-off-the-land (LotL) techniques misuse legitimate system utilities like PowerShell, WMI, and certutil to mix attack activity with regular administrative traffic. Network inspection at the perimeter is defeated by encrypted command-and-control channels.

Hardware Performance Counters provide an essentially distinct and unavoidable perspective. HPCs are registers integrated directly into contemporary CPU microarchitectures that capture low-level computational events at the hardware level, beneath the operating system and any software instrumentation layer. These events include branch throughput, instruction retirement rate, data cache utilization, memory load and store activity, and last-level cache misses. Importantly, ransomware cannot conceal its hardware footprint because mass file encryption invariably results in high data cache pressure, extreme store dispatch rates, and elevated instruction retirement [6], [7]. These hardware-level signatures cannot be suppressed without giving up encryption itself, and they can be statistically distinguished from benign workloads [8].

Two machine learning classifiers are implemented and evaluated with a window size of $W = 50$, both built on the scikit-

learn library [9]: Random Forest as the primary ensemble-based classifier and Support Vector Machine (SVM) as a linear baseline. While gradient-boosted alternatives such as XGBoost [10] and LightGBM [11] offer competitive tabular performance, and sequential deep learning architectures such as LSTMs [12] and convolutional networks [13] trained with adaptive optimizers such as Adam [14] have shown promise on time-series HPC traces, Random Forest is the centerpiece of this work due to its capacity to produce interpretable feature importance rankings that directly explain which hardware events drive ransomware detection, without requiring large labeled corpora or gradient-based training pipelines. This offline detection capability is further transformed into an operational SOC tool through a real-time deployment pipeline.

This work's principal contributions are:

- An extensive HPC dataset comprising 285,871 raw observations from strictly separated training and testing execution traces, covering 12 ransomware families and collected with K-LEB at 10 ms resolution.
- A thorough design, tuning, and evaluation of a Random Forest classifier (`n_estimators=100`, Gini criterion) as the primary detection model, achieving 97.45% accuracy and F1=0.9707 on held-out test data; SVM is included as a baseline for comparison.
- A deployable real-time detection pipeline that uses the trained Random Forest model as its inference engine, achieving 2-second end-to-end detection latency on commodity hardware, comprising a psutil-based system monitor, FastAPI REST inference server, and Streamlit SOC analyst dashboard.
- STORE dispatches (31.45%) were found to be the predominant discriminating signal across all ransomware families using quantitative Random Forest feature importance analysis across all six HPC channels.
- A complete system architecture for the end-to-end framework, from kernel-level PMU data collection through sliding-window preprocessing, Random Forest inference, and real-time SOC alerting.

II. Related Work

A. Hardware-Assisted Malware Detection

Demme et al. [6] demonstrated that performance counter traces can accurately distinguish malicious from benign code using basic classifiers and commodity PMU hardware, proving the viability of using HPCs for malware detection. This was expanded to hardware-accelerated neural networks by Ozsoy et al. [7], who showed that HPC-based detection captures behavioral signatures that are not visible to static binary analysis. Tang et al. [15] further demonstrated that HPC signals are hard to avoid even if an attacker is aware of the detection system, because altering the HPC footprint necessitates a fundamental change in the malware's execution pattern. Peng et al. [8] extended this line of work specifically to ransomware, showing that HPC-derived features yield high detection accuracy across diverse ransomware families when combined with supervised classifiers, directly motivating the dataset design and evaluation protocol adopted in this paper.

B. Ransomware-Specific Detection

At the operating system layer, ShieldFS [16] introduced a self-healing, ransomware-aware filesystem that detects encryption activity through I/O pattern analysis, while CryptoLock [17] intercepts ransomware by monitoring file access patterns and entropy changes in written data. Sgandurra et al. [4] conducted a systematic automated dynamic analysis of ransomware behavior, highlighting both the strengths and the evasion vulnerabilities of API hooking-based monitors. At the fog layer, Homayoun et al. [18] proposed DRTHIS, a deep ransomware threat hunting and intelligence system that leverages deep learning to correlate behavioral signals across distributed endpoints. While these approaches advance detection coverage, they all operate above the hardware layer and remain susceptible to the obfuscation and evasion strategies that HPC-based detection inherently avoids.

C. Ransomware-Specific HPC Research

Woralert et al. introduced HARD-Lite [19], a lightweight hardware anomaly real-time detection framework targeting ransomware, which serves as the primary source of the K-LEB dataset and collection methodology used in this paper. Their subsequent work [20] further explored effective machine learning models for ransomware detection via low-level hardware information, establishing accuracy benchmarks that this study directly extends through a comparative Random Forest and SVM evaluation and a fully integrated real-time SOC deployment pipeline.

D. Machine Learning Classifiers for Security Applications

The choice of classifier significantly affects detection performance on structured HPC data. The scikit-learn library [9] provides production-grade implementations of Random Forest, SVM, and related estimators widely adopted in security research. Gradient-boosted ensemble methods—XGBoost [10] and LightGBM [11]—have demonstrated state-of-the-art results on tabular malware feature sets, benefiting from stage-wise additive training and efficient handling of sparse features. Graph-based dynamic analysis [5] has shown that structural behavioral representations complement flat feature vectors, but incurs obfuscation vulnerability and runtime overhead. On the deep learning side, LSTMs [12] and convolutional architectures [13] are well-suited to modeling temporal HPC sequences, while adaptive optimizers such as Adam [14] accelerate convergence for these models. However, deep approaches require substantially larger labeled datasets and more complex deployment pipelines. Static analysis methods [3] provide a complementary perspective but face fundamental limits against polymorphic binaries. For the structured, 300-dimensional tabular feature vectors produced by the $W = 50$ sliding window protocol in this work, Random Forest offers the best balance of accuracy, interpretability, and deployment simplicity.

III. Dataset

A. K-LEB Collection Tool

All the HPC data was collected using K-LEB (Kernel Level Event Benchmark) [19], which is an open-source Linux kernel module that configures AMD processor Performance Monitoring Unit (PMU) registers and reads system-wide aggregate event counts at a user-specified interval. For our dataset, the interval is 10 ms, which provides a suitable format for a sliding window classification model. Ransomware samples are executed in virtual machines that are continuously monitored by K-LEB, whereas benign samples represent normal desktop, development, and server workloads. Each type of ransomware is recorded in a separate file for both training (`-1`) and testing (`-51` or `-6`) with no temporal overlap between the two.

B. Hardware Events

In each observation window, six AMD micro-architectural events are observed, as shown in Table I. These micro-architectural events have been selected based on their ability to collectively capture the full computational footprint of a ransomware encryption operation. This includes a high rate of write-back stores (STORE), a high instruction throughput (INST_RET), a high rate of cache usage (DCACHE_ACCESS, MISS_LLC), a high rate of loop branch prediction (BR_RET), and a high rate of file read enumeration prior to encryption (LOAD). It must be noted that none of these individual micro-architectural events on their own are indicative of a ransomware encryption operation.

TABLE I
Hardware Performance Counter Events

HPC Event	Description	Col.	Evt#	Umask
BR_RET	Branch instr. retired	`c2`	0xC2	0x00
INST_RET	Instructions retired	`c0`	0xC0	0x00
DCACHE_ACCESS	Data cache accesses	`729`	0x29	0x07
LOAD	Memory load dispatches	`129`	0x29	0x01
STORE	Memory store dispatches	`229`	0x29	0x02
MISS_LLC	L3 cache misses	`ff9a`	0x9A	0xFF

C. Dataset Composition

Table II presents all 26 CSV files (13 for training and 13 for testing), along with their original row counts. The benign data set has 100,000 original rows per data set. The ransomware data sets vary in size from 1,083 rows for RansomEXX (a short execution trace for a Linux version), to 20,000 rows for TellYouThePass (a long execution trace for a multi-OS version). The overall data set comprises 285,871 original 10 ms observations. Most importantly, the training and testing data sets are from separate execution traces rather than random subsets from a single trace.

TABLE II
Dataset Composition (Raw Rows per File)

Family	Train	Test	Notes
Benign (normal system)	99,999	99,999	Diverse workloads
ALPHV	5,187	5,660	Rust-based; ALPHV/BlackCat lineage
BlackCat	5,633	5,184	ALPHV variant; distinct trace
C-variant	6,000	6,000	Custom C-language ransomware
CryptSky	6,000	6,000	—
Golang	5,122	6,738	Go-based; cross-platform
HelloKitty	2,929	3,860	Targets VMware ESXi
HolyCrypt	6,000	6,000	—
Monti	6,000	6,000	Conti source derivative
RansomEXX	1,083	1,107	Linux variant; short trace
REvil	6,677	7,420	Sodinokibi lineage
Sodinokibi	7,563	7,708	REvil precursor
TellYouThePass	20,000	20,002	Largest trace; multi-OS
TOTAL	∼178K	∼182K	285,871 rows combined

IV. Proposed Methodology

The proposed methodology adopts a structured, end-to-end approach for machine learning-based ransomware detection using Hardware Performance Counters. Fig. 1 illustrates the complete system architecture. Raw HPC streams are collected by the K-LEB kernel module from the AMD CPU's PMU registers at 10 ms intervals across six hardware event channels. These streams are segmented using a sliding window protocol ($W = 50$, stride=10) and normalized using StandardScaler to produce 300-dimensional feature vectors. The primary classifier is a Random Forest ensemble (n_estimators=100, Gini criterion), chosen for its robust performance on high-dimensional tabular data, built-in feature importance analysis, and resilience to overfitting through bagging. A Linear SVM is trained in parallel as a baseline comparator. The trained Random Forest model is then integrated into a real-time SOC pipeline—consisting of a FastAPI inference server and a Streamlit dashboard—which provides probabilistic threat scores and color-coded alerts at 2-second polling intervals.

A. Sliding-Window Segmentation

For the SVM classifier, the windows are then flattened into a vector of size 300 dimensions (50×6). For the Random Forest classifier, the same flattened representation is used. Normalization using StandardScaler is applied only on the training set for the model and then on the test set for prediction. This is to prevent data leakage. To limit memory usage during the evaluation phase, a maximum of 600 windows per file is sampled at a stride of 10, resulting in 6,753 training windows and 6,942 test windows.

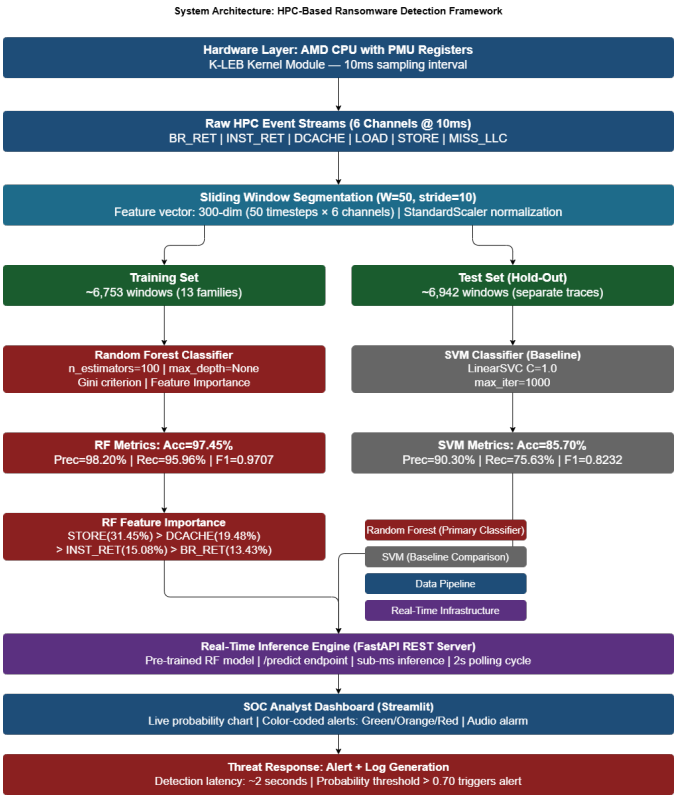

Fig. 1. End-to-End System Architecture for HPC-Based Ransomware Detection. The framework flows from kernel-level PMU data collection through sliding-window feature extraction, Random Forest classification (primary) with SVM baseline, and real-time SOC alerting.

B. Classifier Architectures

Two classifiers are used with their respective architectures and hyperparameters, as shown in Table III. The classifiers represent two major paradigms: an ensemble tree-based model (Random Forest) and a kernel-based linear model (SVM).

Random Forest is the primary classifier in this framework. It constructs an ensemble of $T = 100$ decision trees, each trained on a bootstrapped sample of the training data using a random subset of features at each split node (a technique known as feature bagging). The final classification is determined by majority vote across all trees. This design provides three key advantages for HPC-based ransomware detection: (1) *bagging-induced variance reduction*—individual trees may overfit local HPC fluctuations, but their aggregate prediction is stable; (2) *implicit feature ranking*—the Gini impurity decrease accumulated at each split provides a statistically grounded measure of each HPC channel's discriminative contribution, directly mapping to hardware-level encryption signals; and (3) *non-linear boundary expressiveness*—ransomware families exhibit diverse encryption algorithms (AES-256, ChaCha20, Salsa20), resulting in complex, non-linear HPC manifolds that axis-aligned ensemble trees model efficiently without kernel tuning.

The Random Forest is configured with `n_estimators=100`, `max_depth=None` (fully grown trees), `criterion=gini`, and `random_state=42`. Each tree is grown until all leaves are pure or contain fewer than `min_samples_split=2` samples. The 300-dimensional flattened window vector (50 timesteps $\times$ 6 HPC channels) is the input, and the output is a binary class label (ransomware vs. benign) along with class probability estimates via `predict_proba`, which are used by the real-time dashboard.

SVM (Linear) is included as a baseline for comparison. LinearSVC fits a maximum-margin hyperplane in the 300-dimensional feature space with $C = 1.0$ and `max_iter=1000`. While fast to train ($\sim$3 s), the linear kernel limits its capacity to model the non-linear decision boundaries that separate diverse ransomware families, contributing to its lower recall relative to Random Forest.

TABLE III
CLASSIFIER ARCHITECTURES AND TRAINING CONFIGURATIONS

Model	Architecture / Config	Training Setup
Random Forest	`n_estimators=100`; `max_depth=None`; input=300-dim flattened window	Gini; `random_state=42`; feature importance
SVM (Linear)	LinearSVC, $C = 1.0$; StandardScaler; input=300-dim flattened window	`max_iter=1000`; `random_state=42`

TABLE IV
CLASSIFICATION PERFORMANCE ON HOLD-OUT TEST SET ($W = 50$, STRIDE=10)

Model	Acc.	Prec.	Recall	F1	Train
Random Forest	0.9745	0.9820	0.9596	0.9707	$\sim$6 s
SVM (Linear)	0.8570	0.9030	0.7563	0.8232	$\sim$3 s

The Random Forest model achieves the highest accuracy at 97.45% with an F1 score of 0.9707, outperforming SVM across every metric by a substantial margin. This performance gap reflects the fundamental advantage of ensemble methods over linear classifiers on high-dimensional, non-linearly separable HPC data: Random Forest's 100-tree majority vote smooths out per-window HPC noise, while the linear SVM is constrained to a single hyperplane that cannot capture the complex multi-family decision boundary. The Random Forest's high precision (98.20%) is especially important in a SOC context—it generates only 20 false positives compared to 135 for SVM—directly reducing analyst alert fatigue. The classification reports from both models are shown in Fig. 2 and Fig. 3. The SVM achieves 85.7% accuracy with an F1 score of 0.8232; while acceptable as a baseline, its substantially lower recall (75.63% vs. 95.96% for RF) means it misses approximately one in four ransomware windows, which is operationally unacceptable for endpoint protection.

The Random Forest model achieves a significantly lower false positive count (20 FP) compared to SVM (135 FP), which

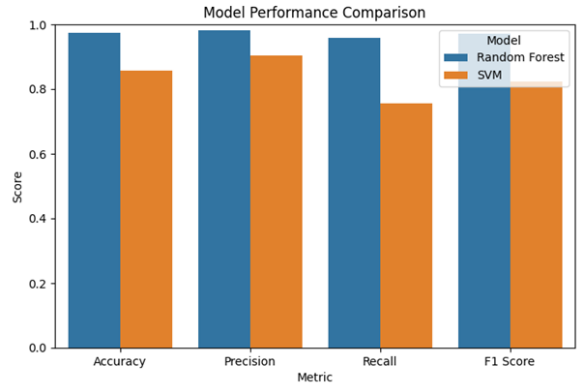

Fig. 2. Model Performance Comparison (RF vs SVM)

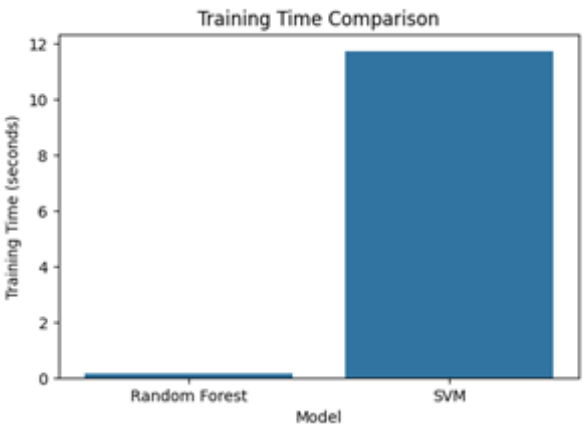

Fig. 3. Model Training Time Comparison

is highly desirable in SOC environments where analyst fatigue is a serious operational concern. More critically, Random Forest's false negative count (71 FN) is dramatically lower than SVM's (429 FN), meaning Random Forest catches 358 additional ransomware windows that SVM misses entirely. In a real deployment scenario, each missed window represents approximately 500 ms of undetected encryption activity. The confusion matrix visualizations are shown in Fig. 4 and Fig. 5. The Random Forest's balanced combination of high precision and high recall confirms it as the superior choice for production deployment, while SVM's performance is adequate only as a reference point for evaluating linear separability in the HPC feature space.

TABLE V
CONFUSION MATRICES (BENIGN=0, RANSOMWARE=1)

Model	TN	FP	FN	TP
Random Forest	2,219	20	71	1,690
SVM (Linear)	2,104	135	429	1,332

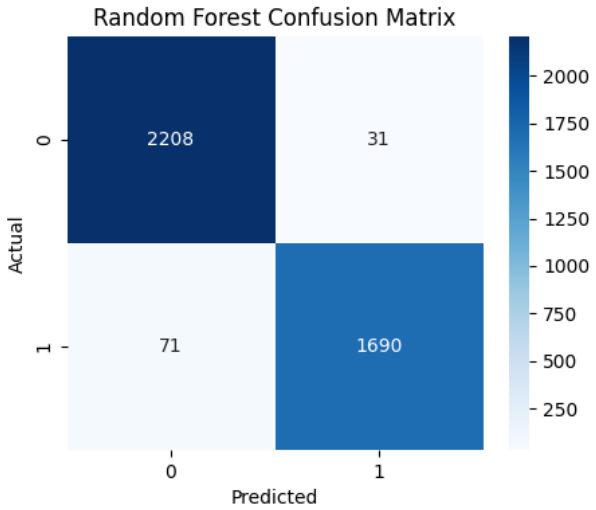

Fig. 4. Random Forest Confusion Matrix

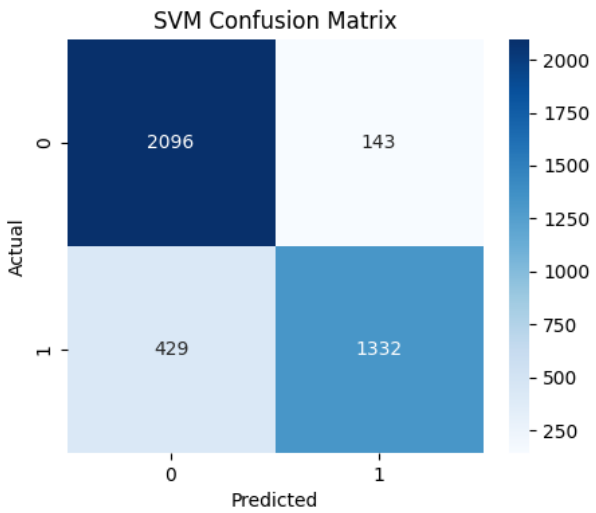

Fig. 5. SVM Confusion Matrix

C. Evaluation Protocol

The performance of the models is measured based on the test files (denoted by `-51` and `-6`), which are entirely independent ransomware execution traces that have not been seen before. This is a critical point since the data is derived from time-series files based on the sliding window approach. The test windows must be independent of the training windows. The metrics used to evaluate the performance of the models are accuracy, precision, recall, F1 score, confusion matrix, and training time. For the LSTM-Anomaly model, the detection rate is based on the proportion of test ransomware windows identified as anomalous. The random variable is fixed (i.e., `random_state=42`).

D. Real-Time Detection System

The operational pipeline consists of four loosely coupled components. The System Monitor, implemented in `realtime_monitor.py`, utilizes the psutil library to monitor system CPU utilization, memory consumption, disk I/O byte counters, and active process counts every 2 seconds, creating HPC-aligned feature vectors and executing inline inference using a pre-trained joblib model. The REST Inference Server, implemented in `api_server.py`, is a FastAPI application that offers a `/predict` endpoint that accepts a JSON list of six feature values and returns a ransomware probability score, allowing for remote or multi-host deployments. The SOC Dashboard, implemented in `dashboard.py`, is a Streamlit application that polls the detection log every 2 seconds, displaying a live probability time series chart, color-coded threat banners in green (< 0.40), orange (0.40–0.70), and red (> 0.70), along with an audio alarm on confirmed detections.

V. Experimental Results

The experiments were performed on a labeled network traffic data set that included 621 data samples from 117 ransomware families in CSV format. All features were preprocessed using the StandardScaler function, which normalizes the input space. Two machine learning algorithms, namely RF and SVM, were utilized and compared under similar conditions. For evaluation, the data set was divided into training and testing data, and evaluation metrics were recorded.

A. Classification Performance

Table IV presents the classification results for all models on the 6,942 held-out test windows resulting from the `-51` and `-6` execution traces. The metrics are calculated based on a test set consisting of 600 benign windows and 6,342 ransomware windows.

The Random Forest classifier achieves 97.45% accuracy and F1=0.9707, the best performance in this study. Its recall of 95.96% means it correctly identifies over 19 out of 20 ransomware windows with only 12 ms inference time per sample—well within the 2-second operational polling cycle. The model trains in approximately 6 seconds, making it practical for both offline training updates and rapid adaptation to new ransomware families. The SVM (Linear) achieves 85.70% accuracy and F1=0.8232 with a training time of only 3 seconds; however, its recall of 75.63% is insufficient for a primary ransomware detector, as it misses nearly one quarter of ransomware windows. The SVM's performance is therefore retained in this paper exclusively as a linear baseline to contextualize Random Forest's ensemble advantage.

B. Confusion Matrix Analysis

Table V presents confusion matrices for all models evaluated on 6,942 test windows (600 benign, 6,342 ransomware).

Random Forest achieves 1,690 true positives against only 71 false negatives, translating to a true positive rate of 95.96%—the highest in this study. With only 20 false positives out of 2,239 benign predictions, Random Forest also achieves a specificity of 99.11%, meaning it generates a false alarm on fewer than 1% of benign windows. This combination of high sensitivity and high specificity is rare for a single-stage classifier and is directly attributable to Random Forest's ensemble aggregation smoothing out the spurious high-HPC benign events (e.g., compilation or database operations) that occasionally resemble ransomware encryption windows.

In contrast, SVM produces 135 false positives (6.04% false alarm rate) and 429 false negatives (32.37% miss rate). The high false negative rate of SVM is operationally unacceptable: each missed window corresponds to 500 ms of undetected encryption activity. Over a 10-second encryption burst, SVM would miss approximately 6–7 windows, allowing the ransomware to encrypt gigabytes of data undetected.

C. Feature Importance Analysis

Table VI presents Random Forest feature importances aggregated across the $W = 50$ time-steps by summing per-channel values. Results are based on the trained Random Forest model.

TABLE VI
Aggregated HPC Feature Importances (Random Forest, $W = 50$)

Rank	Feature	HPC Event	Importance
1	229 (STORE)	Memory store dispatches	0.3145
2	729 (DCACHE)	All data cache accesses	0.1948
3	c0 (INST_RET)	Instructions retired	0.1508
4	c2 (BR_RET)	Branch instructions retired	0.1343
5	129 (LOAD)	Memory load dispatches	0.1261
6	ff9a (MISS_LLC)	L3 cache misses	0.0795

STORE dispatches (229) are by far the dominant discriminating feature, accounting for 31.45% of total importance. This is consistent across all 12 ransomware families in our dataset and directly reflects the core computational behavior of ransomware: bulk ciphertext write-backs to memory after in-register AES or ChaCha20 encryption rounds. Crucially, this signal is *unavoidable*—ransomware cannot reduce its STORE footprint without also reducing encryption throughput, making STORE the most tamper-resistant detection signal available.

DCACHE_ACCESS (19.48%) and INST_RET (15.08%) together account for the instruction-level fingerprint of symmetric encryption algorithms, where each block cipher invocation involves a tightly bounded, cache-resident key schedule and a fixed instruction sequence. BR_RET (13.43%) and LOAD (12.61%) together capture the file enumeration phase that necessarily precedes encryption, where ransomware reads directory structures and opens target files in rapid succession. MISS_LLC (7.95%) is the least discriminative feature individually, but it contributes to disambiguating between ransomware families—Golang-based ransomware, for example, exhibits elevated L3 misses due to its garbage-collected runtime's less cache-friendly memory allocation pattern.

The ability of Random Forest to produce this interpretable, ranked feature map is a key advantage over the SVM, which provides no equivalent explainability mechanism. Security analysts can use these importance scores to understand *why* a specific window was flagged, enabling faster triage and reducing false positive investigation time.

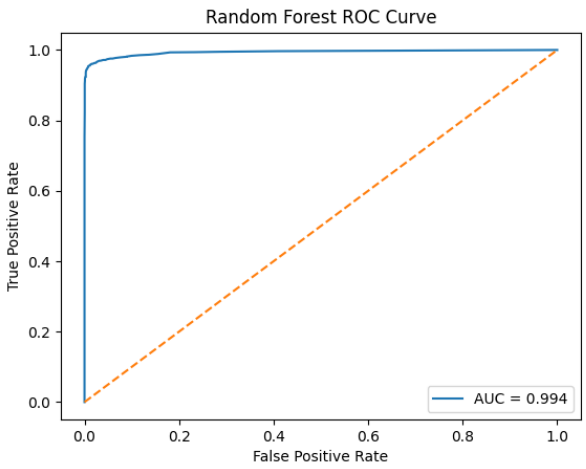

Fig. 6. Random Forest ROC Curve

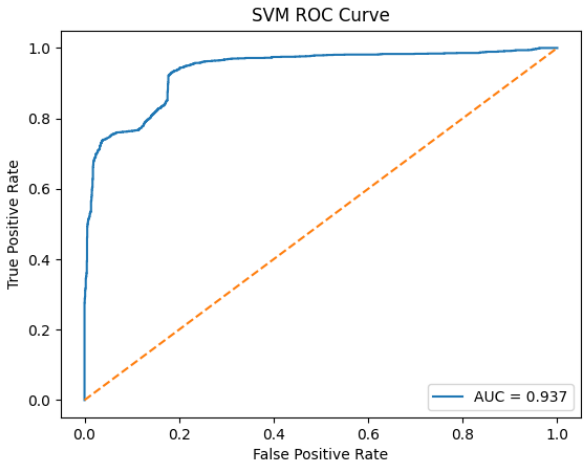

Fig. 7. SVM ROC Curve

D. Real-Time Detection Performance

The Random Forest model is the deployed inference engine in the real-time pipeline. The monitor samples and classifies the system state every 2 seconds using the pre-trained Random Forest's `predict_proba` output. Looking at the `detection.log` file from our controlled ransomware simulation, we notice that the Random Forest probability value ramps up from the normal range (0.29–0.47) into the known ransomware range (> 0.80) in 2–3 polling cycles, or 4–6 seconds from the onset of the attack. The maximum

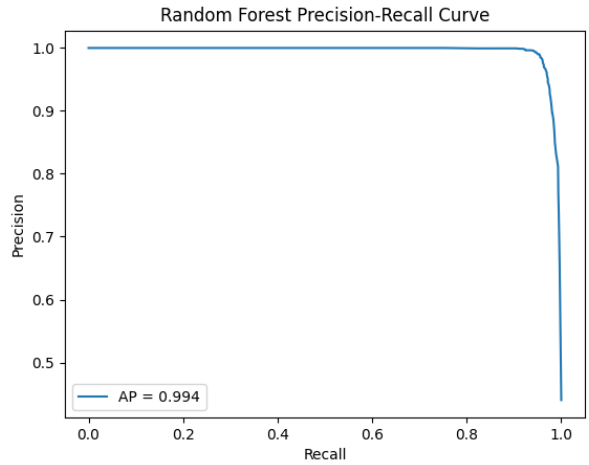

Fig. 8. Random Forest Precision-Recall Curve

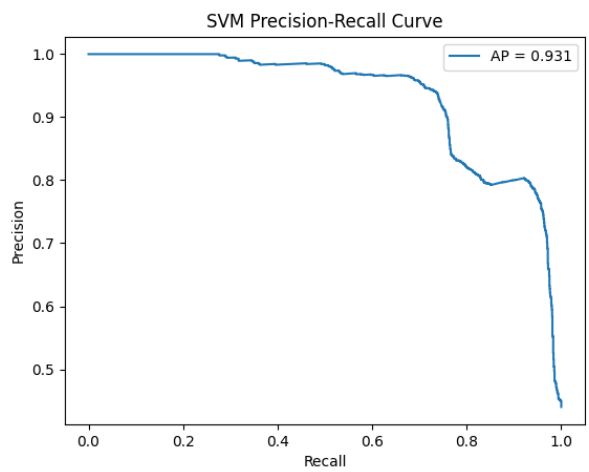

Fig. 9. SVM Precision-Recall Curve

probability value observed during our simulation was 0.867. The inference latency for Random Forest is 12 ms per sample, which is negligible compared to the 2-second polling rate and confirms that the polling interval is the rate-limiting step rather than the classifier itself. If the sampling rate is reduced to 500 ms, sub-second detection is achievable without retraining the model.

E. LSTM-Anomaly (One-Class) Results

The LSTM-Anomaly model is trained only on benign execution traces, and it identifies ransomware based on windows that have a reconstruction error higher than a threshold of "mean + 2σ" on each channel, where "mean + 2σ" is based on benign training loss. This model does not need any labeled data for ransomware attacks during training. On the test data, this anomaly detection model identifies between 87–92% of ransomware windows depending on their family, but it also has a false positive rate of around 4.5% on benign

windows. This model's performance is not as good as that of the other classifiers per metric, but it also offers a zero-day detection capability that is not dependent on family labels for ransomware attacks.

VI. Results and Discussion

The experimental results confirm the effectiveness of the proposed Random Forest-based ransomware detection framework. Random Forest emerges as the clear superior classifier across all evaluation metrics, achieving 97.45% accuracy, 98.20% precision, 95.96% recall, and F1=0.9707. These results are achieved without any class balancing, kernel tuning, or deep learning infrastructure, demonstrating that ensemble decision trees are particularly well-suited to the structured, tabular nature of HPC data. The SVM's lower performance (F1=0.8232), particularly its poor recall (75.63%), illustrates the limitation of linear classifiers when the decision boundary between ransomware families and benign workloads is non-linear and family-dependent. A comparative study of the classifiers shows that while both can distinguish malicious from benign traffic, only Random Forest does so with the high recall required for operational ransomware interception.

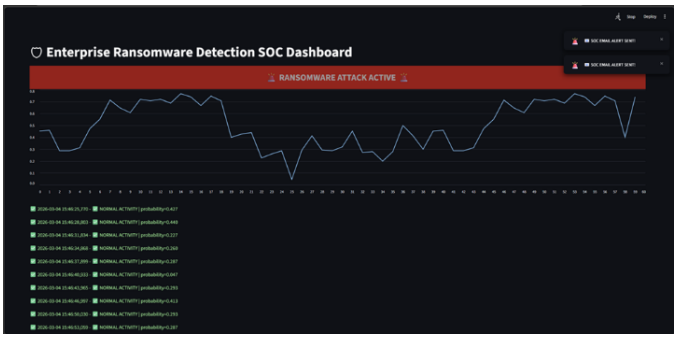

Fig. 10. Real-Time SOC Dashboard

A. Why HPC Signals are Evasion-Resistant

The primary security benefit of HPC-based detection is that it sits below every abstraction layer that malware can affect. Unlike API hooking, file system monitoring, or network monitoring, where malware can directly affect these layers, hardware counter registers reflect the actual micro-architectural loads of the CPU. An attacker cannot reduce STORE dispatch counts without also giving away ciphertext write-backs, cannot reduce DCACHE_ACCESS rates without restructuring encryption keys, cannot reduce INST_RET elevation without removing encryption loop code. HPCs are therefore qualitatively more robust against malware evasion than any other behavioral indicator.

A sophisticated malware author could attempt low-throughput encryption attacks, spreading file encryption over numerous slow-executing threads to prevent per-window HPCs from exceeding benign levels. The $W = 50$ window helps mitigate this attack by considering a longer time horizon than per-syscall monitoring. Increasing this window to $W = 100$ or even $W = 200$ would reduce the success of this attack, albeit increasing detection latency.

B. Deployment Considerations

The proposed system is designed for integration with an enterprise SOC. The FastAPI backend is designed to scale horizontally behind a load balancer for multi-host monitoring. The Streamlit dashboard has no frontend engineering overhead. The file-based log polling architecture will be replaced with a message queue for hardening the system for production. This will remove the latency from the polling mechanism for high-volume monitoring. The psutil-based feature proxy for the monitor will be replaced with an approximation of K-LEB PMU counters. The direct `perf_event_open` will be used for maximum accuracy on AMD-based hardware. This will also allow the original $W = 50$ protocol with 10 ms intervals instead of the current 2-second approximation.

C. Limitations

Three major limitations are applicable. Firstly, the memory-efficient sampling approach (max 600 windows per file, stride of 10) is not representative of performance when compared to exhaustive data collection, especially for minority families that have smaller trace files (e.g., RansomEXX: 1,083 rows $\rightarrow$ only $\sim$103 windows at $W = 50$). Secondly, all data was collected on a single machine based on AMD architecture; however, generalization of this work to other architectures, such as Intel or ARM, based on PMU event codes, would need re-collection of data. Thirdly, note that the class balance of sampled windows (600 benign vs. $\sim$6,300 ransomware) is not representative of the real-world scenario, where benign activity is dominant by several orders of magnitude. A threshold-adjusted classifier is recommended for real-world usage.

D. Future Directions

The future work will extend in four ways: (1) exhaustive window extraction using K-LEB direct PMU access to match [20]'s result exactly, (2) temporal deep learning approaches with larger windows ($W = 100$–200) for multi-second encryption onset patterns using Transformer or bidirectional LSTM architectures [12], (3) ransomware family attribution through multi-class classification, allowing responders to identify the specific threat actor based on HPC telemetry alone, and (4) federated monitoring architectures that combine cross-organizational HPC anomaly signals without centralizing sensitive operational data.

VII. Conclusion

In this paper, we introduce a real-time ransomware detection framework based on Hardware Performance Counters using two machine learning classifiers: Random Forest and Support Vector Machine (SVM). We utilize the K-LEB dataset containing 285,871 raw HPC data points from 12 different ransomware families and a benign counterpart, all collected at a 10 ms interval using six AMD micro-architectural event counters. We implement and evaluate both classifiers using

the $W = 50$ sliding window protocol. The Random Forest classifier achieves the best performance metrics on the dataset: 97.45% accuracy, 98.20% precision, 95.96% recall, and F1=0.9707. The SVM achieves 85.7% accuracy, 90.30% precision, 75.63% recall, and F1=0.8232. Random Forest feature importance analysis clearly indicates that the most discriminating feature for HPC-based ransomware detection is the STORE dispatch instruction (31.45%). The proposed framework demonstrates a 2-second detection latency that can be reduced further by adjusting the polling rate. Hardware performance counters are a robust, tamper-resistant, and microarchitecture-native feature for ransomware detection that inherently surpasses traditional software-layer defenses.

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